BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' : Multiple Choice Questions

Maximum Marks: 33

Note: This section consists of 33 part questions and all are to be answered. Each question carries ONE mark. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Q1. Choose the correct answer for each from the given options:

Q1. 3.01×10^{23} molecules of oxygen gas at STP occupy a volume of:

(A) 22.4 dm^3
(B) 224 dm^3
(C) 11.2 dm^3
(D) 2.24 dm^3

Q2. 602.10 has significant figures:

(A) 3
(B) 4
(C) 5
(D) 6

Q3. One mole of H_2O contains this number of Hydrogen atoms:

(A) 6.02×10^{23}
(B) 1.204×10^{24}
(C) 3.01×10^{23}
(D) 1.204×10^{23}

Q4. The number of molecules in this pair are same:

(A) 10gm H_2 and 10gm CH_4
(B) 10gm H_2 and 50gm CH_4
(C) 10gm H_2 and 80gm CH_4
(D) 10gm H_2 and 16gm CH_4

Q5. The vapour pressure of water at 100°C will be:

(A) 760 torr
(B) 76 torr
(C) 14.2 psi
(D) 100 torr

Q6. Number of crystal system on the basis of unit cell is:

(A) 5
(B) 6
(C) 7
(D) 8

Q7. Capillary action in liquids is due to:

(A) Viscosity
(B) Surface Tension
(C) Density
(D) Fluidity

Q8. The rate of diffusion of CO_2 is equal to:

(A) CH_4
(B) CO
(C) SO_2
(D) C_3H_8

Q9. The (n + l) value for 5d orbital is:

(A) 4
(B) 5
(C) 6
(D) 7

Q10. By emitting α-particles $_{92}U^{238}$ converts into:

(A) $_{82}Pb^{206}$
(B) $_{84}Po^{210}$
(C) $_{89}Ac^{227}$
(D) $_{90}Th^{234}$

Q11. The maximum number of electrons in a particular sub-energy level is:

(A) $2n^2$
(B) n^2
(C) (2l + 1)
(D) 2(2l + 1)

Q12. e/m value is minimum for positive rays when discharge tube contains:

(A) Hydrogen
(B) Helium
(C) Nitrogen
(D) Oxygen

Q13. The bond energy of this molecule is maximum:

(A) CH_4
(B) O_2
(C) N_2
(D) Cl_2

Q14. This molecule has the maximum bond angle:

(A) NH_3
(B) CO_2
(C) SO_2
(D) H_2O

Q15. This hydrogen halide has the highest percentage of ionic character:

(A) HF
(B) HCl
(C) HBr
(D) HI

Q16. This molecule has zero dipole moment:

(A) NH_3
(B) CO_2
(C) H_2O
(D) HCl

Q17. This is an intensive property:

(A) Density
(B) Mass
(C) Mole
(D) Volume

Q18. 1 Cal is equal to:

(A) 0.239 J
(B) 1.98 J
(C) 4.184 J
(D) 8.314 J

Q19. This colour has the shortest wavelength:

(A) Red
(B) Blue
(C) Violet
(D) Orange

Q20. 400 J of heat is supplied to a system at constant volume, the change in internal energy is:

(A) 400 J
(B) -400 J
(C) 300 J
(D) 200 J

Q21. For the reaction $2NH_3 \rightleftharpoons N_2 + 3H_2$ the relation between Kc and Kp is:

(A) Kp = Kc
(B) Kp > Kc
(C) Kp < Kc
(D) Kp < Kc

Q22. The extent of reaction will be maximum for the Kc value:

(A) 10^{-3}
(B) 0.1
(C) 10
(D) 10^3

Q23. In an exothermic reaction increase of temperature favours:

(A) Forward Reaction
(B) Reverse Reaction
(C) to remain in equilibrium
(D) Irreversible Reaction

Q24. Precipitation occur if the product of ionic concentration is:

(A) greater than Ksp
(B) less than Ksp
(C) equal to Ksp
(D) equal to Kc

Q25. When 2 moles of solute are present in $2dm^3$ of solution, then concentration of solution is:

(A) 0.5M
(B) 1M
(C) 2M
(D) 3M

Q26. This one has the highest pH value:

(A) 0.01 M NaOH
(B) 0.02M HCl
(C) 0.01M $NaHCO_3$
(D) 0.1M H_2SO_4

Q27. The oxidation number of Mn in $KMnO_4$ is:

(A) +2 (B) +4

(C) +6 (D) +7

Q28. Universal indicator in acid gives this colour:

(A) Green (B) Yellow

(C) Pink (D) Red

Q29. Human body is an example of:

(A) Open System (B) Closed System

(C) Isolated System (D) Isochoric System

Q30. The reaction $2NO_2 \longrightarrow 2NO + O_2$ has order:

(A) Zero (B) 1st

(C) 2nd (D) 3rd

Q31. These reactions have low value of activation energy:

(A) Slow (B) Fast

(C) Moderate (D) Ionic

Q32. The change in concentration of reacting substance in a unit time is called:

(A) Rate of Reaction (B) Rate Constant

(C) Rate Law (D) Specific Rate Constant

Q33. The threshold energy of a chemical reactions is 31 KJ/mole and the average internal energy of the reactant is 12 KJ/mole the activation energy will be:

(A) 55 KJ/mole (B) 43 KJ/mole

(C) 25 KJ/mole (D) 19 KJ/mole

Section 'B' : Short Answer Questions

Maximum Marks: 32

> Note: Attempt any EIGHT questions from this section. All questions carry equal marks.

Q2.

(i) Define the following: ☆ Avogadro's Number ☆ Heat of Formation ☆ Limiting Reactant ☆ Rate Expression

(ii) Calculate the volume of CO_2 gas produced at 25°C and 740 torr pressure by the combustion of 20 grams of CH_4. $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

(iii) An organic compound on analysis gave the following result: C = 12.8%, H = 2.1%, Br = 85.1%

(iv) What is Ideal Gas? Derive the General Gas equation.

(v) Calculate the density of SO_2 in g/dm³ at 25°C and 300 torr pressure.

(vi) State (n + l) rule and write the electronic configuration of the following: ☆ Mo(Z = 42) ☆ S^{--}(Z = 16) ☆ Sr^{++}(Z = 38)

(vii) Calculate the heat of formation of benzene at 25°C. ☆ $6C + 3H_2 \longrightarrow C_6H_6$; ΔHf = ? ☆ $C + O_2 \longrightarrow CO_2$; ΔHf = -394kJ / mol ☆ $H_2 + ½O_2 \longrightarrow H_2O$; ΔHf = -286kJ / mol ☆ $C_6H_6 + 15/2\ O_2 \longrightarrow 6CO_2 + 3H_2O$; ΔHf = -3267 kJ / mol

(viii) Define Solubility. The solubility of $Mg(OH)_2$ at 25°C is 0.00146gm/dm³. What is the solubility product of $Mg(OH)_2$?

(ix) State and explain Le-Chatelier principle with reference to, the formation of NH_3 by Haber's process or sulphuric acid by contact process.

(x) Define and explain orbital hybridization. Describe sp2 hybridization in C_2H_4.

(xi) Explain any four of the following: ☆ Evaporation is a cooling process. ☆ Milk gets sour more rapidly in summer than in winter. ☆ Glycerine is distilled at reduced pressure. ☆ The ionization potential of nitrogen is more than that of oxygen. ☆ The boiling point of H_2O is greater than HF although electronegativity of fluorine is more than oxygen.

(xii) Write the rate law. Determine the order of reaction using the following data: $A + B \longrightarrow AB$ — Exp.1: [A] = 0.1M, [B] = 0.1M, $dx/dt = 2 \times 10^3\ MS^{-1}$; Exp.2: [A] = 0.2M, [B] = 0.1M, $dx/dt = 8 \times 10^3\ MS^{-1}$; Exp.3: [A] = 0.1M, [B] = 0.2M, $dx/dt = 4 \times 10^3\ MS^{-1}$

(xiii) Define pH. The pH of solution is 8.4. Find: ☆ pOH of the solution ☆ Hydrogen ion concentration ☆ Hydroxyl ion concentration

Section 'C' : Detailed Answer Questions

Maximum Marks: 20

> Note: Attempt any TWO question from this section. All questions carry equal marks.

Q3.

(a) With the help of the experiment of scattering of α-rays, explain the atomic model and its conclusion.

(b) 1 mole of N_2 and 1 mole of O_2 were heated at 2000°C in 10dm³ vessel. Calculate the equilibrium concentration of Nitric Oxide Kc for the reaction is 0.1. $N_2 + O_2 \rightleftharpoons 2NO$

Q4.

(a) Define standard electrode potential. What is standard hydrogen electrode and how is the electrode potential of Zn determined?

(b) Balance any two of the following equations by ion electron method: ☆ $Cr_2O_7^{2-} + I_2 \longrightarrow Cr^{3+} + IO_3^{1-}$ (Acidic medium) ☆ $MnO_4^- + SO_3^{2-} \longrightarrow Mn^{2+} + SO_4^{2-}$ (Acidic medium) ☆ $Br_2 + OH^- \longrightarrow Br^- + BrO_3^- + H_2O$ (Basic medium)

Q5.

(a) Explain the structures of BF_3 and NH_3 on the basis of electron pair repulsion model and hybrid orbital model.

(b) Differentiate any two of the following: ☆ Amorphous solid and Crystalline Solid ☆ Covalent Bond and Coordinate Covalent Bond ☆ Polar and Non-polar Bond ☆ Molarity and Molality

(OR) Define Chemical bond. Explain the ionic bonding in NaCl and discuss its stability.